Answer **all** *the questions*

Each question is followed by **four** *options lettered* A *to* D. *Find the* **correct** *option for* **each** *question and shade* **in pencil** *on your answer sheet, the space which bears the same letter as the option you have chosen. Give only* **one** *answer to* **each** *question. An example is given below.*

If $3n + 2 = 8$, find the value of n.

A. 10
B. 6
C. 3
D. 2

The correct answer is **2**, *which is lettered* **D** *and therefore answer space* **D** *would be shaded.*

⊏A⊐ ⊏B⊐ ⊏C⊐ ■ ⊏E⊐

Think carefully before you shade the answer spaces, erase your first answer completely and shade the appropriate space for the new answer.

Do all rough work in this paper.
Now answer the following questions

1. How long will the simple interest on GH¢ 550.00 at 12 % per annum be GH¢ 132.00?
A. 1 year
B. 2 years
C. 4 years
D. 12 years

2. Expand and simplify: $2(3a + 1) - 3(4a - 3)$.
A. $11 - 5a$
B. $11 - 6a$
C. $11 + 5a$
D. $11 + 6a$

3. The area of a rectangular card is 15 cm^2. If each side of the card is enlarge by a scale factor 3, find the area of the enlarged card.
A. 45 cm^2
B. 75 cm^2
C. 90 cm^2
D. 135 cm^2

4. Factorize: $5ay - by + 15a - 3b$.
A. $(y + 3)(5a - b)$
B. $(y + 5)(3a - b)$
C. $(y - 3)(5a + b)$
D. $(y - 5)(3a + b)$

5. Mr Adu bought 400 bags of maize for his farm animals. If he used 120 bags to feed the animals, find the percentage of the maize left.
A. 70 %
B. 60 %
C. 50 %
D. 40 %

6. A man spends GH¢ 560.00 out of his weekly wage of GH¢ 700.00 and saves the rest. What percent age did he save?
A. 10 %
B. 15 %
C. 20 %
D. 25 %

7. Solve: $3(x - 5) > 15 - 4(8 - x)$.
A. $x < -32$
B. $x < -2$
C. $x < 2$
D. $x < 32$

8. Esi made sales of 15 twenty cedi notes, 14 ten cedi notes and 15 two cedi notes. Find her total sales.
A. GH¢ 305.00
B. GH¢ 440.00
C. GH¢ 470.00
D. GH¢ 740.00

9. When 0.24 is expressed in the lowest form as $\frac{a}{b}$, the denominator is
A. 2.
B. 5.
C. 25.
D. 125.

10. Solve: $4x - 2(x + 5) = -10$.
A. $x = -10$
B. $x = 0$
C. $x = \frac{1}{2}$
D. $x = 2$

11. A rectangular container with dimensions 5 m by 3 m by 4 m is **two-third** full of water. Find the volume occupied by the water in the container.
A. 50 m^3
B. 40 m^3
C. 30 m^3
D. 20 m^3

12. Ama is three times as old as Kofi. The sum of their ages is 40. How old is Ama?
A. 10 years
B. 30 years
C. 37 years
D. 43 years

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13. A point $(-2, 3)$ is reflected in the x-axis. Find the image of the point.
A. $(-3, -2)$
B. $(-3, 2)$
C. $(-2, -3)$
D. $(-2, 3)$

14. A farmer feeds 20 goats with 500 kg of cassava. How many goats can be fed with 200 kg of cassava?
A. 2
B. 5
C. 8
D. 10

15. In an examination, Abu answered **nine** questions in 2 hours. He spent 20 minutes on the first question and the same time on **each** of the remaining questions. How many minutes did he spend on **each** of the **other** questions?
A. 8.0 minutes
B. 10.0 minutes
C. 12.0 minutes
D. 12.5 minutes

16. A trader sold half of a piece of cloth and used two-fifths of the remaining to sew a dress. What fraction of the cloth was left?
A. $\frac{1}{10}$
B. $\frac{3}{10}$
C. $\frac{1}{5}$
D. $\frac{1}{2}$

17. Solve $2^x = 8 \times 2^0$.
A. $x = 3$
B. $x = 2$
C. $x = -2$
D. $x = -3$

18. A boy walked round a circular pond once. If the radius of the pond is 28 m, find the distance covered. [Take $\pi = \frac{22}{7}$]
A. 44 m
B. 88 m
C. 176 m
D. 252 m

19. The points $M(1, 3)$ and $N(4, 5)$ are in the number plane. Find the vector $\overrightarrow{MN}$.
A. $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$
B. $\begin{pmatrix} -3 \\ -2 \end{pmatrix}$
C. $\begin{pmatrix} 5 \\ 8 \end{pmatrix}$
D. $\begin{pmatrix} -5 \\ -8 \end{pmatrix}$

20. Find the rule for the mapping:

1	2	3	4	... t
↓	↓	↓	↓	↓
9	20	31	42	... –

A. $t \rightarrow 10t - 1$
B. $t \rightarrow 8t + 1$
C. $t \rightarrow 11t - 2$
D. $t \rightarrow 7t + 2$

21. If $2y = 5 - 3x$, find x when $y = 1$.
A. $-2\frac{1}{3}$
B. -1
C. 0
D. 1

22. Given that $0.03 \times y = 2.4$, find the value of y.
A. 0.08
B. 0.8
C. 8.0
D. 80.0

23. Find the gradient of the line which passes through the points (2, 3) and (−4, 5).
A. -3
B. $-\frac{1}{3}$
C. $\frac{1}{3}$
D. 3

24. Two interior angles of a triangle are $(3x - 10)^\circ$ and $(4x + 20)^\circ$. Find an expression for the third angle.
A. $(170 - 7x)^\circ$
B. $(150 - 5x)^\circ$
C. $(120 - 7x)^\circ$
D. $(100 - 5x)^\circ$

25. If $\boldsymbol{a} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$ and $\boldsymbol{b} = \begin{pmatrix} -5 \\ -3 \end{pmatrix}$, find $2\boldsymbol{a} - \boldsymbol{b}$.
A. $\begin{pmatrix} 1 \\ 5 \end{pmatrix}$
B. $\begin{pmatrix} -9 \\ -1 \end{pmatrix}$
C. $\begin{pmatrix} 5 \\ 1 \end{pmatrix}$
D. $\begin{pmatrix} -1 \\ -1 \end{pmatrix}$

26. It costs a carpenter GH¢ 25.00 to make a chair. How much should it be sold to make a profit of 40 %.
A. GH¢ 15.00
B. GH¢ 35.00
C. GH¢ 40.00
D. GH¢ 50.00

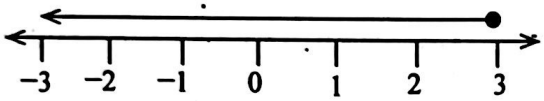
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Turn over

27. Ama has 4 one cedi notes and orders an ice cream for GH¢ 1.75 and two toffees at 50 GP each. How much does she have left?
A. GH¢ 1.25
B. GH¢ 1.75
C. GH¢ 2.25
D. GH¢ 2.75

28. A trader received a commission of 5 % on goods sold at GH¢ 25,000.00. Find the commission.
A. GH¢ 1,250.00
B. GH¢ 1,200.00
C. GH¢ 1,100.00
D. GH¢ 1,000.00

29.



Which of the following inequalities is represented on the number line?
A. $x < 2$
B. $x \leq 3$
C. $x > 3$
D. $x \geq 2$

30. Multiply $(8s - 7)$ by $(8s + 7)$.
A. $64s^2 + 49$
B. $64s^2 - 49$
C. $16s^2 - 42$
D. $16s^2 + 42$

31. The population of a town is 56782. What is this number to three significant figures?
A. 567
B. 568
C. 56700
D. 56800

32. Find the largest value of these numbers: −1, 0, −6, −3.
A. 0
B. −1
C. −3
D. −6

33. A box contains 10 green and 8 white balls of the same size. If a ball is selected at random from the box, what is the probability that it is green?
A. $\frac{1}{10}$
B. $\frac{4}{5}$
C. $\frac{5}{9}$
D. $\frac{4}{9}$

34. Express 36 as a product of primes.
A. 2×3
B. $2^2 \times 3^2$
C. $2^2 \times 3^3$
D. $2^3 \times 3^2$

35. Describe the set of $M = \{2, 3, 5, 7, 11, 13, 17, 19\}$ in words.
A. M = {odd numbers less than 20}
B. M = {factors of 19}
C. M = {prime numbers less than 20}
D. M = {whole numbers less than 20}

36. A survey shows that 28 % of all the men in a village are vegetarian. What is the probability that a man selected at random from the village is a vegetarian?
A. $\frac{7}{25}$
B. $\frac{41}{50}$
C. $\frac{1}{2}$
D. 1

37. Given that $P = \{4, 8, 12, 16, 20\}$ and $Q = \{2, 4, 6, 8, 10\}$, find the product of the members of $(P \cap Q)$.
A. 12
B. 18
C. 24
D. 32

38. Anowa scored an average of 53 in Science and Mathematics. If she scored 50 and 60 in English Language and Social Studies respectively, find her mean score in all the four subjects.
A. 57
B. 56
C. 55
D. 54

39. What is the missing number in the sequence: −5, −2, 1, ... , 7?
A. 2
B. 3
C. 4
D. 5

40. Charles and Helen started a business with an amount of GH¢7,000.00. If their contributions were in the ratio 4:3 respectively, find Helen's contribution.
A. GH¢ 2,500.00
B. GH¢ 3,000.00
C. GH¢ 4,000.00
D. GH¢ 5,000.00

END OF PAPER

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